

Review

Practical Aspects of the Ligand-Binding and Enzymatic Properties of Human Serum Albumin

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Recent work with approaches like recombinant mutants and X-ray crystallography has given much new information about the ligand-binding properties of human serum albumin (HSA). The information increases the understanding of this unique transport and depot protein and could give a structural basis for the possible construction of therapeutic agents with altered HSA-binding properties. A tabulation of high-affinity binding sites for both endogenous and exogenous compounds has been made; it could be useful for the above-mentioned purpose, but it could also be of value when trying to predict potential drug interactions at the protein-binding level. Drug displacement is not always a complication to therapy; it can be used to increase the biological effect of a drug. However, due to rebinding at other sites, the increase in the free concentration of a displaced ligand can be less than expected. Drugs and drug metabolites can also interact covalently with HSA; thiol-containing drugs often bind to the single free cysteine residue of HSA, and glucuronidated drugs react irreversibly with other residues of the protein. Reversible binding of ligands is often stereospecific, and therefore immobilized HSA can be used to separate drug isomers. Albumin-containing dialysates are useful for extracorporeal removal of endogenous toxins and in the treatment of drug overdoses. HSA has different types of hydrolytic activities, which also can be stereospecific. The esterase-like property seems especially useful in converting prodrugs to active drugs in plasma.

Key words high-affinity binding site; mutual displacement; enzymatic property; covalent binding; extracorporeal clearance; immobilized albumin

1. INTRODUCTION

Human serum albumin (HSA) is the most abundant protein in blood plasma and serves as a transport and depot protein for numerous endogenous and exogenous compounds. The protein binding results in an increased solubility in plasma, decreased toxicity, and/or protection against oxidation of the bound ligand. However, binding can also have a significant impact on the pharmacokinetics of drugs in other ways. For example, drug binding to albumin can result in a prolonged *in vivo* half-life of the therapeutic agent. Recently, much new information has been obtained on these interactions at the molecular level, primarily by using recombinant albumin mutants and by X-ray crystallography. That information is reviewed in this paper, which also has a grouping of high-affinity binding sites of albumin. This type of detailed information should shed new light on this unique ligand-binding protein and could provide a structural basis for the possible design of drug derivatives with altered HSA-binding properties. Characterization of the albumin-binding sites could also have more general interest, because they could mimic those of the target cells.

The present grouping of high-affinity binding sites includes drugs and physiological and pathological ligands and should also be useful in predicting possible drug interactions at the albumin-binding level. Mutual interactions between ligands usually result in increased concentrations of the unbound fraction of the ligands in question. However, the effect can be less than expected, because the ligands could be rebound at other sites. Displacement is not always a complication to therapy, but can be used to increase the biological effect of a drug.

Ligands can also bind covalently to HSA, for example,

those possessing a free thiol group. In addition, certain types of drug metabolite interact irreversibly with the protein, and this tendency is especially pronounced for drugs capable of forming acyl glucuronides. The latter type of interaction can be stereoselective.

Reversible binding of ligands to HSA is often stereoselective. This circumstance can be of both analytical and preparative value, because immobilized HSA can be used to separate drug isomers, which often have different biological potencies. Albumin is also of use in extracorporeal removal of endogenous toxins and in treating drug overdoses. In several of the latter methods, albumin can be regenerated online.

HSA is a multifunctional protein and in addition to its unique ligand-binding capabilities it possesses enzymatic properties. The protein has different types of hydrolytic activity, among which the esterase-like property is the most prominent. Recently it has been shown that this activity is useful in converting prodrugs to active drugs.

2. STRUCTURE OF HSA

HSA is a single-chain protein synthesized in and secreted from liver cells. Normally, it is a simple protein, *i.e.*, it lacks prosthetic groups and covalently bound carbohydrate and lipid. The protein has 585 amino acids and a molecular mass of 66500 Da.¹⁾ According to X-ray crystallographic analyses of HSA and its recombinant version (rHSA), the polypeptide chain forms a heart-shaped protein with approximate dimensions of 80×80×80×30 Å and about 67% α -helix but no β -sheet (Fig. 1A).^{2–4)} A recent, combined phosphorescence depolarization-hydrodynamic modeling study proposed that the overall conformation of HSA in neutral solution is very similar to that observed in the crystal form.⁵⁾ X-ray diffraction

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has also shown that the protein has three homologous domains (I–III), and that each of these is comprised of two subdomains (A and B). The A and B subdomains have six and four α -helices, respectively, connected by flexible loops (Fig. 1A). All but one (^{34}Cys) of the 35 cysteine residues are involved in the formation of stabilizing disulfide bonds.

3. LIGAND BINDING

3-1. Binding of Drugs HSA interacts reversibly with a broad spectrum of therapeutic agents. Usually, drugs bind to one or very few high-affinity sites with typical association constants in the range of 10^4 – 10^6 M^{-1} (Table 1). In addition to the primary site(s), several sites of lower or of very low affinity often exist. In the following, attention focuses on high-affinity drug binding, because the *in vivo* concentration of administered drugs is almost always much lower than that of HSA (about 0.6 mM). The pioneering work of Sudlow *et al.*,⁶⁾ which was based on displacement of fluorescent probes, revealed that most drugs bind with high affinity to one of two sites, called site I and site II (Table 1).

3-1-1. Site I Beforehand, it is very difficult, or perhaps impossible, to determine whether a ligand binds with high affinity to this site. However, typical site I ligands seem to be dicarboxylic acids and/or bulky heterocyclic molecules with a negative charge localized in the middle of the molecule. The site must be large, because ligands as big as bilirubin can be bound (see 3-3. **Endogenous Compounds**), and because several examples have been found of independent binding of two different compounds to the site.^{7–10)} Several lines of evidence suggest that the site also is very flexible. (i) The fact that ligands with very different chemical structures bind to the region with high affinity indicates that the site is adaptable.^{1,9)} Recently, this point of view has been supported by the results of Petersen *et al.*,¹¹⁾ who examined the binding of bilirubin to eight different single-residue mutants of rHSA. (ii) Many examples of mutual interactions between bound ligands have been observed. Even though such findings can probably also be explained by partially overlapping binding sites, they are often suggested to be caused by conformational changes in the albumin molecule (allosteric effect or anti-cooperativity). (iii) Single-residue mutations in this region of albumin have a significant effect on the conformational and thermal stability of the protein, much more than mutations in site II.^{12,13)} Thus site I seems to be capacious and flexible and to have a large number of individual ligand-binding sites that sometimes are independent of each other but in other cases influence each other mutually. To characterize the site in more detail, several attempts to subdivide the site have been made.^{8,10,14)}

In the crystal structures of HSA and rHSA, site I is formed as a pocket in subdomain IIA (Fig. 1A) and involves the lone tryptophan of the protein (^{214}Trp).^{4,15)} The inside wall of the pocket is formed by hydrophobic side chains, whereas the entrance to the pocket is surrounded by positively charged residues.⁴⁾ The knowledge of amino acid residues of potential importance for site I has greatly facilitated detailed studies of ligand binding to the site. For example, high-affinity binding of warfarin has been studied using selected rHSA mutants^{13,16)} and by X-ray diffraction of rHSA–warfarin–myristate complexes (Fig. 1B).¹⁷⁾ Both residues contributing posi-

tively (*e.g.*, ^{214}Trp and ^{218}Arg) and residues contributing negatively (*e.g.*, ^{199}Lys and ^{242}His) to binding have been identified. The diffraction study also revealed that the two enantiomers of the drug bind in essentially the same manner to the protein. Such detailed information provides a structural basis for the possible design of warfarin derivatives with altered HSA-binding properties.

3-1-2. Site II Ligands binding to site II (also called the indole-benzodiazepine site) are often aromatic carboxylic acids with a negatively charged acidic group at one end of the molecule away from a hydrophobic center (*e.g.*, non-steroidal antiinflammatory drugs [NSAIDs]). Site II seems to be smaller, or more narrow, than site I, because apparently no large ligands (*e.g.*, bilirubin, hemin, hematin, or other porphyrins) bind to it and because apparently no one has succeeded in subdividing this binding region. It also appears to be less flexible, because binding often is strongly affected by stereoselectivity. The classical example is the finding that L-tryptophan binds with an affinity about 100 times higher than that of the D-isomer.¹⁸⁾ Furthermore, substitution of ligands with a relatively small group can strongly influence binding. For example, diazepam but not its fluorinated analogue flunitrazepam binds to site II,¹⁹⁾ and substitution of a methyl group for the α -hydrogen in L-tryptophan results in blocking of binding.¹⁾ Thus, although the site can bind several ligands

Table 1. High-Affinity Binding of Ligands to Site I and Site II of HSA^{a)}

Site I	$K_1 (\text{M}^{-1})^b$	Site II	$K_1 (\text{M}^{-1})$
Drugs:			
Warfarin ¹⁾	3.4×10^5 ¹⁰⁾	Diazepam ¹⁾	3.8×10^5 ⁴⁸⁾
Azapropazone ¹²⁰⁾	2.8×10^5 ⁹⁾	Ketoprofen ¹³³⁾	2.5×10^6 ¹³⁶⁾
Phenylbutazone ¹⁾	1.5×10^6 ¹⁰⁾	Chlofibrate ¹⁾	7.6×10^5 ¹³⁷⁾
Acenocoumarol ¹⁰⁾	2.2×10^5 ¹⁰⁾	Iopanoate ¹⁾	6.7×10^6 ¹³¹⁾
Salicylate ¹⁾	1.9×10^5 ⁹⁾	Ibuprofen ^{22,68)}	2.7×10^6 ²¹⁾
Indomethacin ¹⁾	1.4×10^6 ¹²⁹⁾	Pirprofen ⁷²⁾	3.9×10^5 ⁷²⁾
Tolbutamide ¹⁾	4.0×10^4 ¹³⁰⁾	S-Naproxen ¹⁾	3.7×10^6 ⁸⁹⁾
Chlorpropamide ¹⁾	3.3×10^5 ⁹⁾	Ethacrynate ¹⁾	1.7×10^6 ³⁴⁾
Iophenoxate ¹⁾	7.7×10^7 ¹³¹⁾	Chlorothiazide ¹³⁴⁾	5.5×10^4 ¹³⁴⁾
Iodipamide ¹⁰⁾	9.9×10^6 ¹⁰⁾	Diclofenac ⁷⁶⁾	3.3×10^6 ⁷⁶⁾
Sulfadimethoxine ¹²¹⁾	9.0×10^4 ¹²¹⁾	S(–)-Thiamylal ⁵⁶⁾	8.7×10^4 ⁵⁶⁾
Sulfathiazole ⁹⁾	2.5×10^4 ⁹⁾	S-Etadolac ⁵⁹⁾	2.0×10^5 ⁵⁹⁾
Furosemide ¹²²⁾	1.9×10^5 ¹²²⁾	Carprofen ⁷⁵⁾	$> 10^6$ ⁷⁵⁾
Tenoxicam ¹²³⁾	3.7×10^5 ¹²³⁾	Suprofen ⁶⁸⁾	N.D.
R-Sulbenicillin ¹¹¹⁾	5.2×10^3 ¹¹¹⁾	Benoxaprofen ²²⁾	N.D.
Valproate ⁶⁰⁾	2.8×10^5 ⁶⁰⁾	Cicloprofen ²²⁾	N.D.
Piretanide ¹²²⁾	9.5×10^4 ¹²²⁾	Azidodeoxythymidine ¹⁵⁾	N.D.
Spiroglactone ¹²⁴⁾	3.0×10^3 ¹²⁴⁾	Norharmane ¹³⁵⁾	N.D.
Canrenoate ¹²⁴⁾	2.0×10^5 ¹²⁴⁾	Propofol ²⁴⁾	N.D.
Quercetin ¹²⁵⁾	2.7×10^5 ¹²⁵⁾	Flurbiprofen ^{22,68)}	N.D.
Benzylthiouracil ¹²⁶⁾	4.1×10^4 ¹²⁶⁾		
Oxyphenbutazone ⁴²⁾	3.5×10^5 ⁴²⁾		
S-Carbenicillin ¹²⁷⁾	2.4×10^3 ¹²⁷⁾		
R-Sulbenicillin ¹²⁷⁾	5.2×10^3 ¹²⁷⁾		
Cosalane ¹²⁸⁾	N.D. ^{c)}		
Endogenous Compounds:			
Bilirubin ¹¹⁾	9.5×10^7 ¹³²⁾	Octanoate ⁴⁸⁾	1.6×10^6 ⁴⁸⁾
Eicosanoids ¹⁾	N.D.	L-Tryptophan ⁴⁸⁾	4.4×10^4 ⁴⁸⁾
3-Carboxy-4-methyl-5-propyl-2-furanpropanoate ⁴⁴⁾	1.3×10^7 ⁴⁴⁾	Monooleoyl glycerol ⁴⁹⁾	4.0×10^5 ⁴⁹⁾
		L-Thyroxine ¹⁾	3.0×10^5 ¹³⁸⁾
		Chloride ion ²¹⁾	7.2×10^2 ¹³⁹⁾
		Indol-3-acetate ⁴⁴⁾	2.1×10^5 ⁴⁴⁾
		Indoxylsulfate ⁴⁴⁾	1.6×10^6 ⁴⁴⁾
		Hippurate ⁴⁴⁾	1.0×10^4 ⁴⁴⁾

a) The table shows some of the ligands that have been proposed to bind with high affinity and $n_1=1$ to either site I or site II. b) K_1 are high-affinity association constants (Scatchard constants) obtained at pH 7.4. c) N.D., not determined.

with high affinity (Table 1), it appears to be more restricted than site I.

Crystallographic analyses have assigned site II to subdomain IIIA (Fig. 1A).^{4,15} Sugio *et al.*⁴ studied the crystal structures of both HSA and rHSA at a 2.5 Å resolution and found that site II is a pocket constructed in principally the same way as site I. Among the individual amino acid residues of this subdomain, ⁴¹⁰Arg and ⁴¹¹Tyr are usually assumed to be important. This is also the case with the binding of, for example, ketoprofen. However, investigations performed with rHSA mutants showed that ⁴¹⁰Arg is not important for diazepam binding.¹²

3-1-3. Additional Sites Although the site I and site II classification of Sudlow *et al.*⁶ remains useful, it cannot account for high-affinity binding of all drugs. Those authors concluded that the strongly bound drugs probenecid, amitriptyline, and debrisoquine do not bind significantly to either of the two sites. Furthermore, the location of the digoxin site has not yet been fully clarified. In addition, it is probably not possible to relate all high-affinity sites to the model of Sudlow *et al.*⁶ if a drug binds to more than two sites. For example, Jakoby *et al.*²⁰ examined the binding of tolbutamide by calorimetric titrations and heteronuclear two-dimensional (2-D) NMR spectroscopy and concluded that the drug binds to three discrete sites with the same or comparable affinities and not to only one high-affinity site, as indicated in Table 1. Two of the sites are reported to be located within sites I and II, whereas the third site most probably is located elsewhere in the albumin molecule.

Even though positively charged drugs mainly interact with α_1 -acid glycoprotein, several of these bind to albumin,^{1,21} although the locations of the primary sites for that type of ligand are still unknown. In this connection, it is of interest to note that Wanwimolruk *et al.*²² have studied the binding of five aliphatic amines with different chain lengths to HSA, and concluded that they do not bind to site II. It is probably also unlikely that they bind to site I. However, the same authors reported²³ that basic NSAIDs bind in their unionized form to site I.

Bhattacharya *et al.*²⁴ described the high-resolution crystal structures of rHSA with propofol or halothane. Propofol binds in subdomains IIIA (site II) and IIIB, with the former site having the highest affinity (Fig. 1C). The latter site most probably also binds anions of long-chain fatty acids with high affinity (see **3-3. Endogenous Compounds**). In contrast, three primary sites were found for halothane, one in subdomain IIIA (site II) and two in a solvent-exposed trough at the outside interface between subdomains IIA and IIB (Fig. 1D). Eckenhoff *et al.*²⁵ using solubilized HSA and rHSA mutants, observed that halothane, isoflurane, and 1-chloro-1,2,2-trifluorocyclobutane bind primarily in the interdomain cleft between IA and IIA (Fig. 1A), which has ²¹⁴Trp as an important binding residue. However, binding of the anesthetics to site II was also evident.

3-1-4. ³⁴Cys The cysteine residue at position 34 in subdomain IA is located in a crevice on the surface of the protein and does not participate in any disulfide bridges. On average, about half of the sulfhydryl groups of isolated HSA are present as mixed disulfides with cysteine or glutathione or are oxidized to sulfenic, sulfinic, or sulfonic states. However, most of the blocked and/or oxidized thiols could be a

result of the isolation procedures used, because in normal fresh plasma these albumin fractions have been reported to be very low, *i.e.*, less than 10%.²⁶ The remaining sulfhydryl groups are freely accessible (mercaptalbumin) and represent the largest fraction of free thiols in blood. Mercaptalbumin and probably also part of albumin forming mixed disulfides are able to bind nitric oxide and metal ions such as Ag⁺, Hg²⁺, and Au⁺ as well as homocysteine²⁷ and several drugs. Among the latter are bucillamine derivatives,²⁸ aurothiomalate,²⁹ auranofin,³⁰ D-penicillamine,^{31,32} captopril,^{32,33} ethacrynate,³⁴ and cisplatin (which also binds to methionine residues).³⁵

Covalent interaction between thiol-containing drugs and ³⁴Cys of albumin usually also takes place *in vivo*. For example, *N*-acetyl-L-cysteine (and L-cysteine) binds to albumin in plasma; the interaction can be quantitated by a high-performance liquid chromatographic (HPLC) method using hexaiodoplatinate.³⁶ Furthermore, D-penicillamine,^{32,37} captopril,^{32,33} meso-2,3-dimercaptosuccinate,³⁸ and SA3786 (a bucillamine derivative)³⁹ bind to albumin in the serum of patients receiving these drugs. Interestingly, the reactivity in patient sera is much higher than that found in albumin solutions³³ and in sera of healthy volunteers.³⁹

3-2. Nondrug Probes Modification of ligands with relatively simple substituents can have a major influence on binding. Therefore it is usually not possible to predict which site(s) a new drug will bind to with high affinity. However, a rapid and simple screening can be performed using probes. Most often fluorescent probes are used, and a large decrease in the fluorescence of a bound probe indicates a common binding site. Among the conditions for this approach are that the probe should bind to only one site of albumin, or that the primary and secondary binding constants are very different. If the latter is the case, the probe-to-protein molar ratios should be kept well below unity to minimize secondary probe binding. Examples of site I probes, in addition to warfarin, are 5-dimethylaminonaphthalene-1-sulfonamide (DNSA),⁶ dansylamide,^{40,41} dansyl-L-glutamine,⁴⁰ dansyl-L-asparagine,¹⁰ dansyl-L-lysine,⁴² *n*-butyl *p*-aminobenzoate,¹⁰ prodan,⁴³ and phenol red,⁷ and among the site II probes are dansylsarcosine,⁶ dansyl-L-proline,⁴⁰ dansylglycine,⁴¹ and 7-alkylaminocoumarin-4-acetic acids.⁴⁴ Care should be exercised with respect to site I especially, because, as mentioned above, several examples of independent co-binding of two ligands to this site have been reported.

3-3. Endogenous Compounds HSA has multiple binding sites of varying affinities for nonesterified fatty acid anions, and the locations of the sites for many of the saturated ones (decanoate, dodecanoate, myristate, palmitate, and stearate) in the crystal structure of rHSA are now known in great detail.⁴⁵ Palmitate, which is the quantitatively most important saturated fatty acid anion under physiological conditions, can bind to seven sites (Fig. 2A). Most sites are within subdomains, but some are formed by adjacent subdomains or by residues on the surface of the protein. So far, the latter two types of binding site have not been reported for non-aliphatic ligands. The sites are located as follows. One molecule of palmitate, and of other long-chain fatty acid anions, can bind very weakly to site I, and two molecules bind with higher affinity to subdomain IIIA, which also houses site II. However, the site of the highest affinity is most probably the

one formed inside subdomain IIIB.^{1,21,45,46} The three remaining sites are placed in the center of subdomain IB, between subdomains IA and IIA, and in a shallow trench on the surface of the protein at the interface between subdomains IIA and IIB.⁴⁵ Recently, it was found that the *cis*-forms of oleate and arachidonate bind to the same seven sites of rHSA, and that in each site the overall configuration of the saturated and unsaturated fatty acid anions is broadly similar.⁴⁷ The situation for anions of medium-chain fatty acids such as octanoate seems to be somewhat different. Binding of octanoate has not yet been studied by X-ray crystallography, but the results of competition studies with solubilized HSA suggest that octanoate has one high-affinity binding site in site II.⁴⁸ High-affinity binding of monoacylglycerides, such as monooleoylglycerol, also occurs at site II.⁴⁹

Of the amino acids, only L-tryptophan (site II) and cysteine (to ³⁴Cys) bind to HSA. In contrast to simple, inorganic cations, only a few of the simple, inorganic anions bind to HSA, for example, Cl⁻ which binds to site II. Bilirubin has been reported to bind to a dynamic and flexible site in subdomain IIA.¹¹ In contrast, the primary binding site for hemin is located in domain I.⁵⁰ Several pieces of evidence suggest that eicosanoid binding occurs at site I¹ and that L-thyroxine appears to bind to site II. The binding sites for bile acids are still not known. For example, lithocholate and its glucuronide and sulfate have been reported to have a common high-affinity binding site, which is different from those of warfarin, diazepam, digitoxin, and bilirubin.⁵¹

In addition to physiological substances, HSA can bind some pathological compounds such as uremic toxins with high affinity. 3-Carboxy-4-methyl-5-propyl-2-furanpropanoate (CMPF) binds to site I,^{44,52} whereas indol-3-acetate, indoxylsulfate, and hippurate all bind to site II.⁴⁴

3-4. Drug-drug Interactions Because albumin has a limited number of regions for high-affinity binding and because the protein is a flexible molecule, binding of a drug often influences simultaneous binding of other drugs. Information about such influences is important, because alteration in protein binding may alter the volume of distribution, clearance, and elimination of a drug and may modulate its therapeutic effect. Therefore it is useful to try to classify drug-binding sites so that the risk of drug interactions can be evaluated. Table 1 is such an attempt. However, although a classification is useful, it should be used with care, because in addition to competitive binding, drugs, as well as other ligands, can influence simultaneous binding of another drug by electrostatic effects and/or binding-induced conformational changes in the protein. When evaluating the effect of protein binding on drug therapy, it is important to take into account not only potential drug-drug interactions but also the fact that drug metabolites can bind to albumin with high affinity; many examples of both conditions can be found in the literature (for reviews, see refs. 1, 31, and 53).

Drug-drug interactions at the protein-binding level are usually regarded as problematic secondary effects. However, such interactions can also be useful for therapeutic purposes. For example, in the nephrotic syndrome furosemide and other loop diuretics have only a small effect on diuresis. This diuretic resistance is most probably caused by drug binding to albumin in the urine, but coadministration of a displacer such as bucolome significantly increases the diuretic effect of

furosemide.⁵⁴ Drug displacement can also be useful in non-pathological conditions. Technetium-99m-labeled mercaptoacetylglycylglycylglycin binds strongly to albumin but displacement by coadministered bucolome increases its unbound concentration and thereby makes it more useful for renal scintigraphy.⁵⁵ The same study showed that 6-methoxy-2-naphthylacetate (a major metabolite of nabumetone) renders, by principally the same mechanism, *N*-isopropyl-*p*-[¹²³I]iodoamphetamine better suited for cerebral imaging. In addition, under these conditions the two radiolabeled test compounds are eliminated much faster from the body. Other benefits of the displacement method with radiolabeled ligands are shorter waiting times and lower radiation doses for the patients.⁵⁵

3-5. Drug-endogenous Compound Interactions

Drug binding can be modulated by simultaneous binding of endogenous compounds. Anions of long-chain fatty acids can at low ligand-to-protein molar ratios ($\leq 2:1$), most probably allosterically, diminish binding of *S*(-)- and *R*(+)-thiamylal,⁵⁶ diazepam,²¹ L-tryptophan,²¹ and dansylsarcosine^{57,58} to site II. The effect on etodolac binding is dependent on the stereochemistry of the drug, because binding of the (*R*)-form was increased, whereas binding of the (*S*)-form was decreased.⁵⁹ At higher relative concentrations, the fatty acid anions often displace site II drugs by competition. For site I ligands, the situation usually is different. At molar ratios of $\leq 3-4:1$, anions of long-chain fatty acids increase binding of, for example, furosemide (in serum samples^{60,61}), warfarin,²¹ DNSA,⁵⁷ and bilirubin.²¹ In contrast, at molar ratios of $\geq 5-6:1$, ligand binding is competitively diminished by the fatty acid anions.

Anions of medium-chain fatty acids, *e.g.*, octanoate, bind with high affinity to site II (Table 1), which is why a pronounced inhibitory effect of these ligands on the binding of site II drugs is expected. This has also been reported to be the case with diazepam,⁴⁸ ibuprofen,²² flurbiprofen,²² thiamylal,⁵⁶ dansylsarcosine,⁵⁸ and L-tryptophan.⁴⁸ Octanoate only binds to site I with low affinity. Therefore the displacing effect on site I drugs is probably very small.

The improving effect of long-chain fatty acid anions on the binding of certain site I ligands was originally explained by induction of conformational changes in the albumin molecule rendering site I more suitable for ligand binding. However, recent X-ray diffraction studies have proposed a supplementary or alternative mechanism. In the absence of fatty acids, albumin binds 2,3,5-triiodobenzoate (TIB) at sites I and II.¹⁵ In the presence of myristate, TIB becomes displaced from site II but rebinds at a new, myristate-induced binding site in subdomain IB³ (Fig. 2B). Perhaps a similar phenomenon could partly or totally explain the above-mentioned enhancements in binding of site I-ligands.

Several drugs have a strong displacing effect on bilirubin binding. This has been found to be the case with valproate,⁶² ceftriaxone,⁶³ sulfonamides such as sulfisoxazole,⁶⁴ and the following drugs⁶⁵: oxyphenbutazone, phenylbutazone, sulfinpyrazone, ketophenylbutazone, glibenclamide, tolbutamide, warfarin, salicylate, and furosemide. Because of their bilirubin-displacing effect, it has been recommended that these drugs should not be given to, for example, neonates in danger of kernicterus. It has been suggested that all drugs used for newborns should be tested for bilirubin-displacing

properties.⁶³) In contrast to the above examples, diazepam and seven other benzodiazepines have no effect on high-affinity bilirubin binding.⁶⁶

Chronic renal failure is associated with accumulation of uremic toxins in serum, among which are four organic acids (see **3-3. Endogenous Compounds**). The presence of these toxins results in impaired binding of drugs and other ligands to HSA. Examples of displaced ligands are diazepam, warfarin, salicylate, furosemide, L-tryptophan, L-thyroxine (see ref. 44) and phenytoin.⁶⁷) In recent years it has been possible to single out some individual effects of the toxins. Thus CMPF (site I) competitively displaces the site I ligands warfarin,⁶⁸) furosemide,⁶¹) and bilirubin.⁵²) Likewise, the site II toxins indoxyl sulfate, indole-3-acetate, and hippurate compete with the site II probe dansylsarcosine.^{68,69}) Interestingly, indoxyl sulfate is able to displace dansyl-L-asparagine anticooperatively even though the latter binds to site I. In contrast, the toxin has no effect on warfarin binding.⁶⁹)

Chloride ions diminish the binding of warfarin,⁷⁰) tolmetin,⁷¹) pirofen,⁷²) etodolac,⁵⁹) azosemide,⁷³) bilirubin,⁷⁴) and perhaps other ligands as well. The displacing effect of chloride ions is also relevant for *in vitro* binding studies, because some widely used buffers contain a high concentration of chloride ions (for example in the form of 0.9% NaCl), whereas other buffers lack the ion.

3-6. Rebinding after Displacement Alterations in the HSA binding of drugs may lead to a change in the pharmacokinetic properties of these drugs. This risk is especially relevant if two or more drugs compete for the same high-affinity binding site. However, in certain cases the increase in unbound drug concentration is much less than expected, because a major part of the displaced drug rebinds at another site. For example, most of the carprofen that has been competitively displaced from site II by ibuprofen becomes rebound at site I; quantitatively the effect is stereospecific.⁷⁵) Such a site II-to-site I displacement has also been observed for diclofenac in the presence of ibuprofen.⁷⁶)

4. COVALENT BINDING OF DRUG METABOLITES

The formation of acyl glucuronides is a major metabolic pathway for many compounds with a carboxylic acid function. This type of metabolite is a reactive electrophilic species and can therefore, in addition to reversible binding, react covalently with HSA both *in vitro* and *in vivo*. The reactivity seems to depend on the degree of substitution of the carbon adjacent to the aglycone carboxylate function, with the less substituted being more reactive.⁷⁷) The binding can affect the metabolic fate and clearance of the drugs in question. In addition, it can modify the structural and functional properties of the protein.⁷⁸) Furthermore, it is probable that the protein adducts cause toxic effects such as hypersensitivity,^{79,80}) idiosyncratic liver toxicity,⁸⁰) and acute renal toxicity.⁸¹)

Among the drugs that become glucuronidated and interact covalently with HSA are furosemide,⁷⁷) salicylic acid,⁸²) gemfibrozil,⁸³) clofibric acid,⁸⁴) fenofibric acid,⁸⁴) and NSAIDs like tolmetin,^{77,85}) benoxaprofen,⁸⁰) zomepirac,⁷⁷) beclorubic acid,⁷⁷) ibuprofen,⁸⁶) ibufenac,⁸⁶) suprofen,⁸¹) ketoprofen,⁸⁷) and diflunisal.⁸⁸) Stereospecificity was observed for, e.g., carprofen,⁷⁷) fenoprofen,⁷⁷) and naproxen⁸⁹); in all cases the

R-forms exhibited the highest covalent binding. Photoactivated naproxen and frusemide, as well as photoactivated frusemide glucuronide, also bind covalently to HSA.⁷⁸)

Two mechanisms have been proposed for these covalent interactions.^{80,88,90}) One is a displacement mechanism in which a nucleophilic group (NH₂, OH, or SH) of the protein attacks the acyl carbon of the aglycone, displacing the glucuronic acid and forming an amide, ester, or thioester bond between the protein and the drug (a transacylation mechanism). The other involves migration of the acyl group from position 1 to the 2, 3, or 4 position of the sugar, ring-chain tautomerism of the pyranose ring, and, finally, condensation of the aldehyde group of the ring-opened tautomer with a lysine ϵ -amino group on the protein to form an imine (a glycation mechanism). Subsequent Amadori rearrangement of the imine can give a more stable ketoamine derivative. Thus, in the latter mechanism, glucuronic acid is not displaced but instead acts as a bridge between the protein and the drug moiety. In addition to the two mechanisms of covalent interaction, HSA can act like an enzyme, because the protein is able to catalyze the hydrolysis of acyl glucuronides back to the parent drugs^{87,88}) (see 7. ENZYMATIC PROPERTIES).

Tryptic digestion of HSA-metabolite adducts followed by HPLC separation and tandem mass spectrometric analyses of the protein fragments revealed that ¹⁵⁹Lys and ¹⁹⁹Lys are major binding sites for benoxaprofen glucuronide⁸⁰) and tolmetin glucuronide,⁸⁵) respectively. Several other lysine residues as well as serine and arginine residues were identified as less prominent binding sites; most of these are in subdomain IIA and domain III.

Many drugs can themselves bind covalently to HSA. This type of interaction usually involves the cysteine residue in position 34 (see **3-1-4. ³⁴Cys**).

5. MOLECULAR BIOCHROMATOGRAPHY

Enantiomeric forms of a drug can differ in potency, toxicity, and behavior in biological systems. Therefore many chiral analytical and preparative methods have been developed for the separation of drug enantiomers. Often HPLC procedures are used, which are based on albumin bound to agarose activated with cyanogen bromide¹) or immobilized on silica by adsorption⁹¹) or by covalent bonding with glutaraldehyde,^{91,92}) 1,1-carbonyldiimidazole,⁹¹) polymethacryloyl chloride,⁹³) epoxide,⁹⁴) or *N,N*-disuccinimidyl carbonate.⁹⁴) Recently, Zhang *et al.*⁹⁴) have published a mild, quick, and reproducible method in which albumin is bonded to silica with 2,4,6-trichloro-1,3,5-triazine. Usually, the protein is coupled to the matrix by one or more of its amino groups, but, for example, carboxyl and sulphur groups can also be involved. The separation of chiral compounds can be improved by altering the composition or pH of the mobile phase, its flow rate, the amount of immobilized protein, and/or by changing the temperature at which the separation is performed.^{91–95}) Using such methods, it has been possible to separate enantiomers of tryptophan^{92,94,95}) and several other organic compounds^{91,92,94}) including drugs. Among the drugs are oxazepam,⁹³) nine of a series of 12 chiral arylcarboxylic acids,⁹⁶) and fenoprofen, ketoprofen, atropin, benzoin, and warfarin.⁹⁴)

The ability of immobilized albumin to separate drug enan-

tiomers has also been used in other techniques. For example, Nakamura *et al.*⁹⁷⁾ have made a porous hollow-fiber membrane covered by a multilayer of albumin crosslinked with glutaraldehyde. Another, and apparently more widely used, approach, is affinity capillary electrophoresis. This method has been applied to many racemic drugs, and the subject has recently been reviewed in detail by Haginaka.⁹⁸⁾

6. EXTRACORPOREAL CLEARANCE

Albumin can be used to clear the body of endogenous toxins. For example, liver failure and renal failure associated with jaundice and excretory insufficiency usually result in accumulation of various endogenous toxins which, in turn, have been shown to induce hepatocellular apoptosis and necrosis and thereby create a vicious cycle of the disease. Among the toxins are albumin-binding compounds such as bilirubin, bile acids, and uremic toxins and they contribute substantially to the development of multiorgan dysfunction such as those that may be involved in cardiovascular, kidney, and cerebral impairment. In these clinical situations the toxins are usually removed from the patient by hemodialysis, but often the treatment has only a minor effect. Recently, this approach has been improved using thinner dialysis membranes with larger pore size and by adding albumin to the dialysate. In this method, albumin can be regenerated online by passing the dialysate through a charcoal column.⁹⁹⁾

Another albumin-based extracorporeal detoxifier that has been developed recently is the molecular adsorbent recirculating system (MARS). MARS is a cell-free, liver assistance method for the selective removal of albumin-binding substances. It also enables the removal of excess water and water-soluble substances *via* an inbuilt dialysis step.¹⁰⁰⁾ Technically, MARS is a modified and expanded dialysis system using an albumin-containing dialysate that is recirculated and regenerated online by dialysis against a bicarbonate-buffered solution, followed by passage through charcoal and anion-exchanger columns. This method of removing albumin-binding toxins and water-soluble substances was found to result in decreases in hepatic encephalopathy, in increases in mean arterial pressure, and in improvements in kidney and liver function.^{101,102)}

Acute liver failure may be the first manifestation of Wilson disease, in which copper metabolism is impaired. Liver transplantation is usually the preferred treatment if copper continues to accumulate in the patient's body. However, albumin dialysis in the form of a routine continuous veno-venous hemodiafiltration using a 44 g/l albumin-containing dialysate and a slow dialysate flow rate (1–2 l/h) has been shown to be an effective treatment for fulminant Wilson disease, resulting in the removal of protein-binding toxins such as copper and bilirubin.¹⁰³⁾

Endotoxins such as lipopolysaccharides and cytokines, and possibly other toxins, can be removed by passing blood through columns with immobilized albumin.¹⁰⁴⁾

The above-mentioned methods, or modifications thereof, can also be used in the treatment of drug-overdoses.¹⁰⁵⁾ In such cases, however, charcoal hemoperfusion is also useful.¹⁰⁶⁾

7. ENZYMATIC PROPERTIES

In some cases the interaction between a small molecule and albumin results in enzymatic activity. Thus albumin has an enolase activity toward dihydrotestosterone, and the active site for this property has been assigned to the amino-terminal end of the molecule.¹⁰⁷⁾ Albumin is also able to act as a thioesterase, because it has a free sulfhydryl group at ³⁴Cys (Fig. 3). For example, the protein can degrade disulfiram (Antabuse), a clinically important process.¹⁰⁸⁾

HSA can catalyze dehydration of prostaglandin D₂¹⁰⁹⁾ and dehydration and subsequent isomerization of 15-keto-prostaglandin E₂¹¹⁰⁾; the latter reactions were inhibited by phenylbutazone and warfarin but not by several other ligands tested. Amino acid residues of site I can also degrade sulbenicillin, with the *R*-isomer being degraded much faster than the *S*-isomer.¹¹¹⁾ Thus the enzymatic properties of HSA can, like the ligand binding properties, be stereospecific. Among other possible catalytic properties of subdomain IIA (Fig. 1A) is an esterase reaction, which results in transfer of the acetyl group from acetylsalicylic acid to ¹⁹⁹Lys.¹⁾ This lysine residue is also important for rapid trinitrophenylation of HSA and for penicilloxylation of the protein.¹¹²⁾ The readiness with which ¹⁹⁹Lys can attack certain molecules through nucleophilic substitution processes is most probably caused by its location close to ¹⁹⁵Lys in the HSA structure (Fig. 3). Recent restrained molecular dynamic simulations have suggested that the enzymatic activity is facilitated by the simultaneous presence of a deprotonated ¹⁹⁹Lys and a protonated ¹⁹⁵Lys.¹¹²⁾ Among the evidence supporting this model is that it results in the shortest distance between the two residues (6.73 Å), a distance which is in agreement with that determined by X-ray crystallography. It is also in agreement with the experimental finding, that the pK_a of ¹⁹⁹Lys is unusually low, namely *ca.* 8.

Subdomain IIIA of HSA possesses a well-known esterase-like activity toward substrates such as *p*-nitrophenyl acetate¹²⁾ and several *N*-carbobenzoxy-D(L)-alanine *p*-nitrophenyl esters.¹¹³⁾ Residues in this part of HSA are probably also able to detoxify cyanide by reaction with elemental sulfur to form thiocyanate.¹¹⁴⁾ In contrast, assigning hydrolysis of drug glucuronides to HSA regions seems to be more difficult, because the catalytic activity depends on the chemical structure and stereochemistry of the aglycone. Some (*e.g.*, diflunisal glucuronide⁸⁸⁾) become hydrolyzed at site I, others (*e.g.*, carprofen glucuronide¹¹⁵⁾) at site II, and still others (*e.g.*, ketoprofen glucuronide¹¹⁶⁾) by amino acid residues located elsewhere in HSA.

The esterase-like activity of HSA in subdomain IIIA is due to the close proximity of ⁴¹⁰Arg and ⁴¹¹Tyr (Fig. 3); in the crystal structure of the protein the distance between the reactive group of the tyrosine residue and the nitrogen atoms of the arginine residue is only 2.7 Å.¹⁵⁾ The hydrolytic activity toward *p*-nitrophenyl acetate has been studied in detail using several rHSA mutants including ⁴¹¹Ser, and the results indicate that the presence of a hydroxy group in position 411 and the nature of its immediate surroundings are crucial for the esterase activity of HSA.¹²⁾

The enzyme-like properties of HSA can be of practical use. Thus a sensitive kinetic method for quantitating the protein utilizing its hydrolytic activity has been developed using

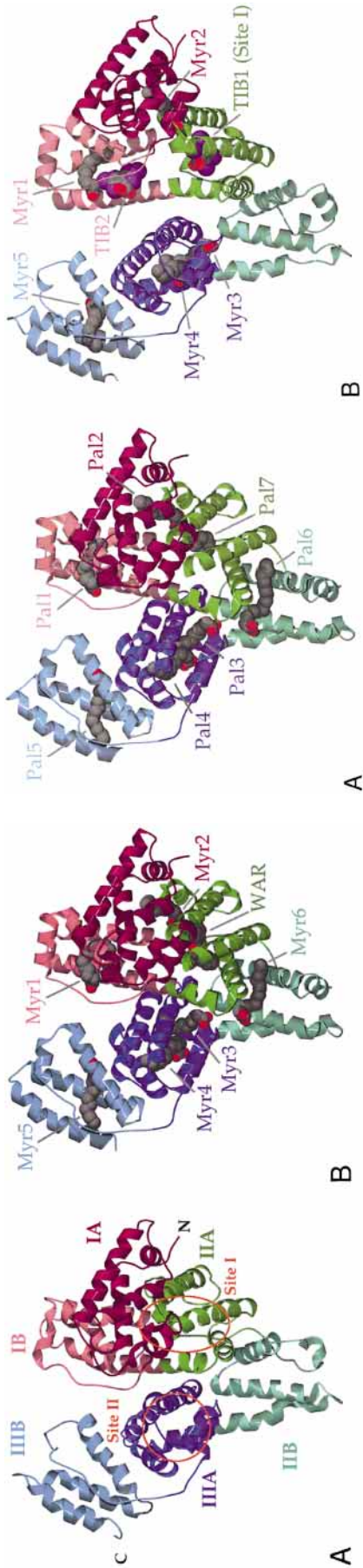


Fig. 1. Crystal Structure of rHSA and rHSA-Ligand Complexes

(A) The subdivision of rHSA (and HSA) into domains (I-III) and subdomains (I-II) is indicated, and the approximate locations of Sudlow's site I and site II are shown. N and C represent the N-terminus and C-terminus, respectively. (B) Binding of the *R*-(+) enantiomer of warfarin (WAR) to crystals of rHSA complexed with six myristate molecules (Myr) (two in subdomain IIIA).²⁷ (C) rHSA complexed with two molecules of propofol (PR).²⁴ (D) rHSA with three molecules of halothane (HAL).²⁴ The illustrations were made with Molscript on the basis of the atomic coordinates 1e7a (B), 1e7a (C), and 1e7b (D) available at the Brookhaven Protein Data Bank.

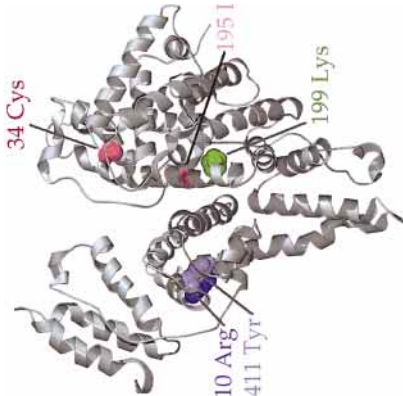


Fig. 2. Crystal Structure of rHSA-Ligand Complexes

(A) Structure of rHSA complexed with seven palmitate molecules (Pal) (two in subdomain IIIA).⁴⁵ (B) Binding of two TIB molecules to crystals of rHSA complexed with five Myr (two in subdomain IIIA).³ The illustrations were made with Molscript on the basis of the atomic coordinates 1e7h (A) and 1bke (B) available at the Brookhaven Protein Data Bank.

Fig. 3. Locations of Five Residues of Importance for the Enzymatic Properties of HSA

The illustration was made with Molscript on the basis of the atomic coordinates 1uor available at the Brookhaven Protein Data Bank.

7-lauroyl-3-(5-methyl-7-sulfonato-2-benzoxazolyl)-coumarin as a substrate; this particular substrate is suitable for both photometric and fluorimetric assays.¹¹⁷⁾

From the pharmaceutical and clinical points of view it is interesting that the esterase-like activity of HSA can be used to activate prodrugs. Human plasma can, with a half-life of a few seconds, activate by hydrolysis olmesartan medoxomil to the active drug olmesartan.¹¹⁸⁾ HSA accounted for about 40% of the total plasma activity, and the esterase-like domain was assigned to subdomain IIIA of which ⁴¹¹Tyr should be important. In contrast to rat and mouse plasma, human plasma and HSA could not activate temocaprin, which is another antihypertensive prodrug.¹¹⁸⁾ Thus prodrug activation varies with animal species and type of prodrug. However, this kind of prodrug activation can be clinically relevant, because although the enzymatic activity of a single HSA molecule is low, the concentration of the protein in the circulation is very high.

The enolase activity of HSA can also be of clinical use. Intracellular accumulation of albumin has been observed in the cytosol of benign and malignant human breast tumors and in mammary tumors of rodents induced by carcinogens.¹¹⁹⁾ Benign and malignant breast tumors can be differentiated on the basis of their cellular albumin contents, because cells of the former type of tumor have a much higher concentration of cytosolic albumin. In addition, both serum and cytosolic HSA possesses an intrinsic enolase activity, converting the 3-keto form of dihydrotestosterone to the 3-enol form, and this activity is much lower in cytosol from malignant tumors. The reduced activity is partly due to the lower albumin concentration and partly due to polymerization and/or binding of the protein to other intracellular proteins. The combined use of the two parameters, intracellular albumin concentration and enolase activity, offers a possibility of reliable differentiation between benign and malignant breast tumors.¹⁰⁷⁾

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REFERENCES

- Peters T., Jr., "All about Albumin: Biochemistry, Genetics, and Medical Applications," Academic Press, San Diego, 1996.
- Carter D. C., Ho J. X., *Adv. Protein Chem.*, **45**, 153—203 (1994).
- Curry S., Mandelkow H., Brick P., Franks N., *Nat. Struct. Biol.*, **5**, 827—835 (1998).
- Sugio S., Kashima A., Mochizuki S., Noda M., Kobayashi K., *Protein Eng.*, **12**, 439—446 (1999).
- Ferrer M. L., Duchowicz R., Carrasco B., García de la Torre J., Acuña A. U., *Biophys. J.*, **80**, 2422—2430 (2001).
- Sudlow G., Birkett D. J., Wade D. N., *Mol. Pharmacol.*, **11**, 824—832 (1975).
- Kragh-Hansen U., *Biochem. J.*, **209**, 135—142 (1983).
- Kragh-Hansen U., *Biochem. J.*, **225**, 629—638 (1985).
- Kragh-Hansen U., *Mol. Pharmacol.*, **34**, 160—171 (1988).
- Yamasaki K., Maruyama T., Kragh-Hansen U., Otagiri M., *Biochim. Biophys. Acta*, **1295**, 147—157 (1996).
- Petersen C. E., Ha C.-E., Harohalli K., Feix J. B., Bhagavan N. V., *J. Biol. Chem.*, **275**, 20985—20995 (2000).
- Watanabe H., Tanase S., Nakajou K., Maruyama T., Kragh-Hansen U., Otagiri M., *Biochem. J.*, **349**, 813—819 (2000).
- Watanabe H., Kragh-Hansen U., Tanase S., Nakajou K., Mitarai M., Iwao Y., Maruyama T., Otagiri M., *Biochem. J.*, **357**, 269—274 (2001).
- Fehske K. J., Schlafer U., Wollert U., Muller W. E., *Mol. Pharmacol.*, **21**, 387—393 (1982).
- He X. M., Carter D. C., *Nature* (London), **358**, 209—215 (1992).
- Petersen C. E., Ha C.-E., Harohalli K., Park D. S., Bhagavan N. V., *Chem. Biol. Interact.*, **124**, 161—172 (2000).
- Petitpas I., Bhattacharya A. A., Twine S., East M., Curry S., *J. Biol. Chem.*, **276**, 22804—22809 (2001).
- McMenamy R. H., Oncley J. L., *J. Biol. Chem.*, **233**, 1436—1447 (1958).
- Chuang V. T. G., Otagiri M., *Biochim. Biophys. Acta*, **1546**, 337—345 (2001).
- Jakoby M. G. IV., Covey D. F., Cistola D. P., *Biochemistry*, **34**, 8780—8787 (1995).
- Kragh-Hansen U., *Pharmacol. Rev.*, **33**, 17—53 (1981).
- Wanwimolruk S., Birkett D. J., Brooks P. M., *Mol. Pharmacol.*, **24**, 458—463 (1983).
- Wanwimolruk S., Birkett D. J., Brooks P. M., *Biochem. Pharmacol.*, **41**, 148—151 (1991).
- Bhattacharya A. A., Curry S., Franks N. P., *J. Biol. Chem.*, **275**, 38731—38738 (2000).
- Eckenhoff R. G., Petersen C. E., Ha C.-E., Bhagavan N. V., *J. Biol. Chem.*, **275**, 30439—30444 (2000).
- Christodoulou J., Sadler P. J., Tucker A., *FEBS Lett.*, **376**, 1—5 (1995).
- Sengupta S., Chen H., Tagawa T., DiBello P. M., Majors A. K., Büdy B., Ketterer M. E., Jacobsen D. W., *J. Biol. Chem.*, **276**, 30111—30117 (2001).
- Narazaki R., Hamada M., Harada K., Otagiri M., *Pharm. Res.*, **13**, 1317—1321 (1996).
- Pedersen S. M., *Ann. Rheum. Dis.*, **45**, 712—717 (1986).
- Ni Dhubbghaill O. M., Sadler P. J., Tucker A., *J. Am. Chem. Soc.*, **114**, 1118—1120 (1992).
- Schaeffer N., Shaw C. F., Thompson H. O., Satre R. W., *Arthritis Rheum.*, **23**, 165—171 (1980).
- Keire D. A., Mariappan S. V., Peng J., Rabenstein D. L., *Biochem. Pharmacol.*, **46**, 1059—1069 (1993).
- Narazaki R., Harada K., Sugii A., Otagiri M., *J. Pharm. Sci.*, **86**, 215—219 (1997).
- Bertucci C., Nanni B., Raffaelli A., Salvadori P., *J. Pharm. Biomed. Anal.*, **18**, 127—136 (1998).
- Ivanov A. I., Christodoulou J., Parkinson J. A., Barnham K. J., Tucker A., Woodrow J., Sadler P. J., *J. Biol. Chem.*, **273**, 14721—14730 (1998).
- Harada K., Naito S., Kawauchi Y., Ishikawa K., Koshitani O., Hiraoka I., Otagiri M., *Anal. Biochem.*, **290**, 251—259 (2001).
- Joyce D. A., Day R. O., *Br. J. Clin. Pharmacol.*, **30**, 511—517 (1990).
- Maiorino R. M., Akins J. M., Blaha K., Carter D. E., Aposhian H. V., *J. Pharmacol. Exp. Ther.*, **254**, 570—577 (1990).
- Narazaki R., Otagiri M., *Pharm. Res.*, **14**, 351—353 (1997).
- Kasai S., Horie T., Mizuma T., Awazu S., *J. Pharm. Sci.*, **76**, 387—392 (1987).
- Muller N., Lapique F., Drelon E., Netter P., *J. Pharm. Pharmacol.*, **46**, 300—304 (1994).
- Matsushita Y., Gouda H., Tsujishita H., Hirono S., *J. Pharm. Sci.*, **87**, 379—386 (1998).
- Moreno F., Cortijo M., González-Jiménez J., *Photochem. Photobiol.*, **69**, 8—15 (1999).
- Irikura M., Takadate A., Goya S., Otagiri M., *Chem. Pharm. Bull.*, **39**, 724—728 (1991).
- Bhattacharya A. A., Grüne T., Curry S., *J. Mol. Biol.*, **303**, 721—732 (2000).
- Curry S., Brick P., Franks N. P., *Biochim. Biophys. Acta*, **1441**, 131—140 (1999).
- Petitpas I., Grüne T., Bhattacharya A. A., Curry S., *J. Mol. Biol.*, **314**, 955—960 (2001).
- Kragh-Hansen U., *Biochem. J.*, **273**, 641—644 (1991).
- Thumser A. E. A., Buckland A. G., Wilton D. C., *J. Lipid Res.*, **39**, 1033—1038 (1998).
- Dockal M., Carter D. C., Rüker F., *J. Biol. Chem.*, **274**, 29303—29310 (1999).
- Takikawa H., Sekiya Y., Yamanaka M., Sugiyama Y., *Biochim. Biophys. Acta*, **1244**, 277—282 (1995).
- Tsutsumi Y., Maruyama T., Takadate A., Goto M., Matsunaga H.,

- Otagiri M., *Pharm. Res.*, **16**, 916—923 (1999).
- 53) Kragh-Hansen U., *Dan. Med. Bull.*, **37**, 57—84 (1990).
 - 54) Uryu Y., Tsutsumi Y., Takamura N., Maruyama T., Suenaga A., Otagiri M., "Abstracts from the International Symposium on Serum Albumin & α_1 -Acid Glycoprotein," Kumamoto, Japan, 2000, p. 98.
 - 55) Kawai K., Takamura N., Nishi R., Jinnouchi S., Nagamachi S., Tamura S., Arimori K., Otagiri M. "Proceedings of the International Symposium on Serum Albumin & α_1 -Acid Glycoprotein," ed. by Otagiri M., Sugiyama Y., Testa B., Tillement J.-P., Kumamoto, Japan, 2001, pp. 181—192.
 - 56) Sueyasu M., Fujito K., Shuto H., Mizokoshi T., Kataoka Y., Oishi R., *Anesth. Analg.*, **91**, 736—740 (2000).
 - 57) Birkett D. J., Myers S. P., Sudlow G., *Mol. Pharmacol.*, **13**, 987—992 (1977).
 - 58) Menke G., Wörner W., Kratzer W., Rietbrock N., *Naunyn-Schmiedeberg's Arch. Pharmacol.*, **339**, 42—47 (1989).
 - 59) Mignot I., Presle N., Lapique F., Monot C., Dropsy R., Netter P., *Chirality*, **8**, 271—280 (1996).
 - 60) Takamura N., Shinozawa S., Maruyama T., Suenaga A., Otagiri M., *Biol. Pharm. Bull.*, **21**, 174—176 (1998).
 - 61) Takamura N., Maruyama T., Otagiri M., *Clin. Chem.*, **43**, 2274—2280 (1997).
 - 62) Yu H. Y., Shen Y. Z., *J. Formos. Med. Assoc.*, **98**, 201—204 (1999).
 - 63) Brodersen R., Robertson A., *Mol. Pharmacol.*, **36**, 478—483 (1989).
 - 64) Brodersen R., *J. Clin. Invest.*, **54**, 1353—1364 (1974).
 - 65) Maruyama K., Harada S., Nishigori H., Iwatsuru M., *Chem. Pharm. Bull.*, **32**, 2414—2420 (1984).
 - 66) Brodersen R., Sjödin T., Sjöholm I., *J. Biol. Chem.*, **252**, 5067—5072 (1977).
 - 67) Mabuchi H., Nakahashi H., *Nephron*, **48**, 310—314 (1988).
 - 68) Sakai T., Takadate A., Otagiri M., *Biol. Pharm. Bull.*, **18**, 1755—1761 (1995).
 - 69) Sakai T., Yamasaki K., Sako T., Kragh-Hansen U., Suenaga A., Otagiri M., *Pharm. Res.*, **18**, 520—524 (2001).
 - 70) Wilting J., van der Giesen W. F., Janssen L. H. M., *Biochem. Pharmacol.*, **30**, 1025—1031 (1981).
 - 71) Matsuyama K., Sen A. C., Perrin J. H., *J. Pharm. Pharmacol.*, **39**, 190—195 (1987).
 - 72) Otagiri M., Masuda K., Imai T., Imamura Y., Yamasaki M., *Biochem. Pharmacol.*, **38**, 1—7 (1989).
 - 73) Lee S. H., Lee M. G., *Biopharm. Drug. Dispos.*, **16**, 615—626 (1995).
 - 74) Weisiger R. A., Ostrow J. D., Koehler R. K., Webster C. C., Mukerjee P., Pascolo L., Tiribelli C., *J. Biol. Chem.*, **276**, 29953—29960 (2001).
 - 75) Rahman M. H., Maruyama T., Okada T., Imai T., Otagiri M., *Biochem. Pharmacol.*, **46**, 1733—1740 (1993).
 - 76) Yamasaki K., Rahman M. H., Tsutsumi Y., Maruyama T., Ahmed S., Kragh-Hansen U., Otagiri M., *AAPS PharmSciTech* **1**: article 12 (<http://www.pharmscitech.com>) (2000).
 - 77) Benet L. Z., Spahn-Langguth H., Iwakawa S., Volland C., Mizuma T., Mayer S., Mutschler E., Lin E. T., *Life Sci.*, **53**, 141—146 (1993).
 - 78) Mizuma T., McDonagh A. F., Lin E. T., Benet L. Z., *Br. J. Clin. Pharmacol.*, **48**, 79—87 (1999).
 - 79) Worrall S., Dickinson R. G., *Life Sci.*, **56**, 1921—1930 (1995).
 - 80) Qiu Y., Burlingame A. L., Benet L. Z., *Drug Metab. Dispos.*, **26**, 246—256 (1998).
 - 81) Smith P. C., Liu J. H., *Xenobiotica*, **23**, 337—348 (1993).
 - 82) Dickinson R. G., Baker P. V., King A. R., *Biochem. Pharmacol.*, **47**, 469—476 (1994).
 - 83) Sallustio B. C., Fairchild B. A., Pannall P. R., *Drug Metab. Dispos.*, **25**, 55—60 (1997).
 - 84) Grubb N., Weil A., Caldwell J., *Biochem. Pharmacol.*, **46**, 357—364 (1993).
 - 85) Ding A., Zia-Amirhosseini P., McDonagh A. F., Burlingame A. L., Benet L. Z., *Drug Metab. Dispos.*, **23**, 369—376 (1995).
 - 86) Castillo M., Smith P. C., *Drug Metab. Dispos.*, **23**, 566—572 (1995).
 - 87) Dubois N., Lapique F., Maurice M.-H., Pritchard M., Fournel-Gigleux S., Magdalou J., Abiteboul M., Siest G., Netter P., *Drug Metab. Dispos.*, **21**, 617—623 (1993).
 - 88) Williams A. M., Dickinson R. G., *Biochem. Pharmacol.*, **47**, 457—467 (1994).
 - 89) Bischer A., Zia-Amirhosseini P., Iwaki M., McDonagh A. F., Benet L. Z., *J. Pharmacokinet. Biopharm.*, **23**, 379—395 (1995).
 - 90) Zia-Amirhosseini P., Ding A., Burlingame A. L., McDonagh A. F., Benet L. Z., *Biochem. J.*, **311**, 431—435 (1995).
 - 91) Harada K., Yuan Q., Nakayama M., Sugii A., *J. Chromatogr. A*, **740**, 207—213 (1996).
 - 92) Tao W. A., Gilpin R. K., *J. Chromatogr. Sci.*, **39**, 205—212 (2001).
 - 93) Millot M. C., Sébille B., Mangin C., *J. Chromatogr. A*, **776**, 37—44 (1997).
 - 94) Zhang Q., Zou H., Wang H., Ni J., *J. Chromatogr. A*, **866**, 173—181 (2000).
 - 95) Yang J., Hage D. S., *J. Chromatogr. A*, **766**, 15—25 (1997).
 - 96) Andrisano V., Booth T. D., Cavrini V., Wainer I. W., *Chirality*, **9**, 178—183 (1997).
 - 97) Nakamura M., Kiyohara S., Saito K., Sugita K., Sugo T., *J. Chromatogr. A*, **822**, 53—58 (1998).
 - 98) Haginaka J., *J. Chromatogr. A*, **875**, 235—254 (2000).
 - 99) Abe Ta., Abe To., Ageta S., Kakuta T., Suzuki N., Hirata H., Shouno M., Saio H., Akizawa T., *Ther. Apher.*, **5**, 58—63 (2001).
 - 100) Mitzner S. R., Stange J., Klammt S., Peszynski P., Schmidt R., Noldge-Schomburg G., *J. Am. Soc. Nephrol.*, **12** (Suppl. 17), S75—S82 (2001).
 - 101) Mitzner S. R., Stange J., Klammt S., Risler T., Erley C. M., Bader B. D., Berger E. D., Lauchart W., Peszynski P., Freytag J., Hickstein H., Loock J., Lohr J. M., Liebe S., Emmrich J., Korten G., Schmidt R., *Liver Transpl.*, **6**, 277—286 (2000).
 - 102) Novelli G., Rossi M., Pretagostini R., Poli L., Peritore D., Berloco P., Di Nicuolo A., Iappelli M., Cortesini R., *Transplant. Proc.*, **33**, 1942—1944 (2001).
 - 103) Kreyman B., Seige M., Schweigart U., Kopp K. F., Classen M., *J. Hepatol.*, **31**, 1080—1085 (1999).
 - 104) Zimmermann M., Busch K., Kuhn S., Zeppezauer M., *Clin. Chem. Lab. Med.*, **37**, 373—379 (1999).
 - 105) Stange J., Mitzner S., Ramlow W., Gliesche T., Hickstein H., Schmidt R., *ASAIO J.*, **39**, M621—M625 (1993).
 - 106) Kawasaki C. I., Nishi R., Uekihara S., Hayano S., Otagiri M., *Am. J. Kidney Dis.*, **35**, 323—326 (2000).
 - 107) Drmanovic Z., Voyatzi S., Kouretas D., Sahpazidou D., Papageorgiou A., Antonoglou O., *Anticancer Res.*, **19**, 4113—4124 (1999).
 - 108) Agarwal R. P., Phillips M., McPherson R. A., Hensley P., *Biochem. Pharmacol.*, **35**, 3341—3347 (1986).
 - 109) Fitzpatrick F. A., Wynalda M. A., *J. Biol. Chem.*, **258**, 11713—11718 (1983).
 - 110) Fitzpatrick F. A., Liggett W. F., Wynalda M. A., *J. Biol. Chem.*, **259**, 2722—2727 (1984).
 - 111) Fleury F., Ianoul A., Berjot M., Feofanov A., Alix A. J. P., Nabiev I., *FEBS Lett.*, **411**, 215—220 (1997).
 - 112) Díaz N., Suárez D., Sordo T. L., Merz, K. M., Jr., *J. Med. Chem.*, **44**, 250—260 (2001).
 - 113) Kurono Y., Kushida I., Tanaka H., Ikeda K., *Chem. Pharm. Bull.*, **40**, 2169—2172 (1992).
 - 114) Jarabak R., Westley J., *J. Biochem. Toxicol.*, **6**, 65—70 (1991).
 - 115) Georges H., Presle N., Buronfosse T., Fournel-Gigleux S., Netter P., Magdalou J., Lapique F., *Chirality*, **12**, 53—62 (2000).
 - 116) Dubois-Presle N., Lapique F., Maurice M.-H., Fournel-Gigleux S., Magdalou J., Abiteboul M., Siest G., Netter P., *Mol. Pharmacol.*, **47**, 647—653 (1995).
 - 117) Gürakar A. R., Wolfbeis O. S., *Clin. Chim. Acta*, **172**, 35—46 (1988).
 - 118) Ikeda T., "Proceedings of the International Symposium on Serum Albumin & α_1 -Acid Glycoprotein," ed. by Otagiri M., Sugiyama Y., Testa B., Tillement J.-P., Kumamoto, Japan, 2001, pp. 173—180.
 - 119) Bhat H. K., Vadgama J. V., Liehr J. G., *Int. J. Oncol.*, **13**, 1087—1091 (1998).
 - 120) Fehske K. J., Muller W. E., Wollert U., *Biochem. Pharmacol.*, **30**, 687—692 (1981).
 - 121) Otagiri M., Nakamura H., Maruyama T., Imamura Y., Takadate A., *Chem. Pharm. Bull.*, **37**, 498—501 (1989).
 - 122) Takamura N., Haruta A., Kodama H., Tsuruoka M., Yamasaki K., Suenaga A., Otagiri M., *Pharm. Res.*, **13**, 1015—1019 (1996).
 - 123) Brée F., Urien S., Nguyen P., Tillement J. P., Steiner A., Vallat-Mollet C., Testa B., Visy J., Simonyi M., *J. Pharm. Pharmacol.*, **45**, 1050—1053 (1993).
 - 124) Takamura N., Maruyama T., Ahmed S., Suenaga A., Otagiri M., *Pharm. Res.*, **14**, 522—526 (1997).
 - 125) Boulton D. W., Walle U. K., Walle T., *J. Pharm. Pharmacol.*, **50**,

- 243—249 (1998).
- 126) Zatón A., Martínez A., de Gandarias J. M., *Biochem. Pharmacol.*, **37**, 3127—3131 (1988).
- 127) Tsuda Y., Tsunoi T., Watanabe N., Ishida M., Yamada H., Itoh T., *Chirality*, **13**, 236—243 (2001).
- 128) Kuchimanchi K. R., Ahmed M. S., Johnston T. P., Mitra A. K., *J. Pharm. Sci.*, **90**, 659—666 (2001).
- 129) Montero M. T., Pouplana R., Valls O., Garcia S., *J. Pharm. Pharmacol.*, **38**, 925—927 (1986).
- 130) Vallner J. J., *J. Pharm. Sci.*, **66**, 447—465 (1977).
- 131) Mudge G. H., Desbiens N., Stibitz G. R., *Drug Metab. Dispos.*, **6**, 432—439 (1978).
- 132) Brodersen R., “Bilirubin, Volume I. Chemistry,” ed. by Heirwegh K. P. M., Brown S. B., CRC Press, Boca Raton, Florida, 1982, pp. 75—123.
- 133) Chuang V. T. G., Kuniyasu A., Nakayama H., Matsushita Y., Hirono S., Otagiri M., *Biochim. Biophys. Acta*, **1434**, 18—30 (1999).
- 134) Takamura N., Rahman M. H., Yamasaki K., Tsuruoka M., Otagiri M., *Pharm. Res.*, **11**, 1452—1457 (1994).
- 135) Fenerty C. A., Lindup W. E., *Biochem. Pharmacol.*, **41**, 1589—1594 (1991).
- 136) Rahman M. H., Yamasaki K., Shin Y. H., Lin C. C., Otagiri M., *Biol. Pharm. Bull.*, **16**, 1169—1174 (1993).
- 137) Meisner H., Neet K., *Mol. Pharmacol.*, **14**, 337—346 (1978).
- 138) Kragh-Hansen U., Minchiotti L., Brennan S. O., Sugita O., *Eur. J. Biochem.*, **193**, 169—174 (1990).
- 139) Scatchard G., Yap W. T., *J. Am. Chem. Soc.*, **86**, 3434—3438 (1964).